SkillClaw: Collective Skill Evolution for Multi-User LLM Agent Ecosystems

From static skills to continuous, evidence-driven, shared improvement

Source: arXiv:2604.08377

Agent Skills Remain Static After Deployment — Users Rediscover Solution

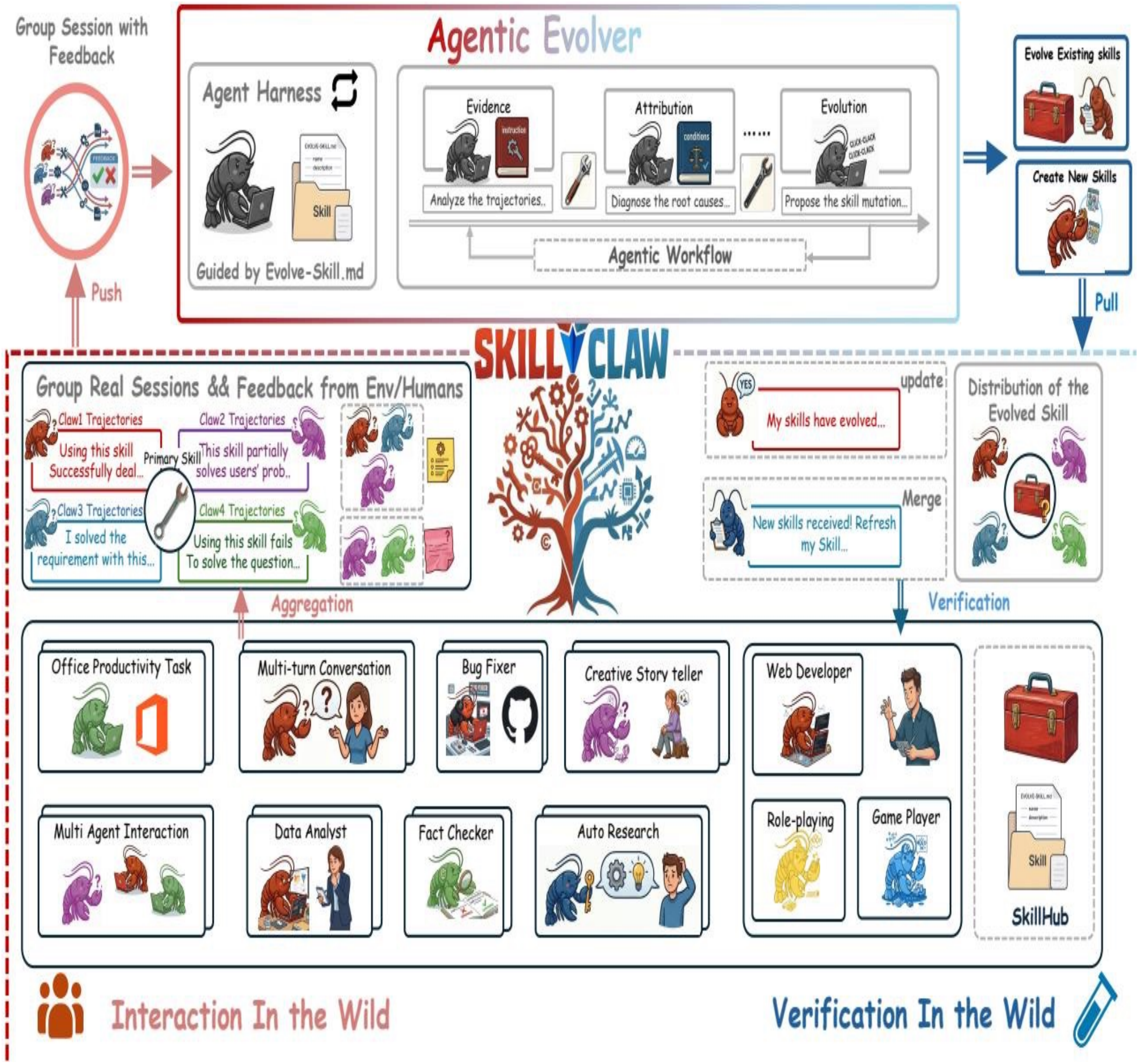
Static paradigm

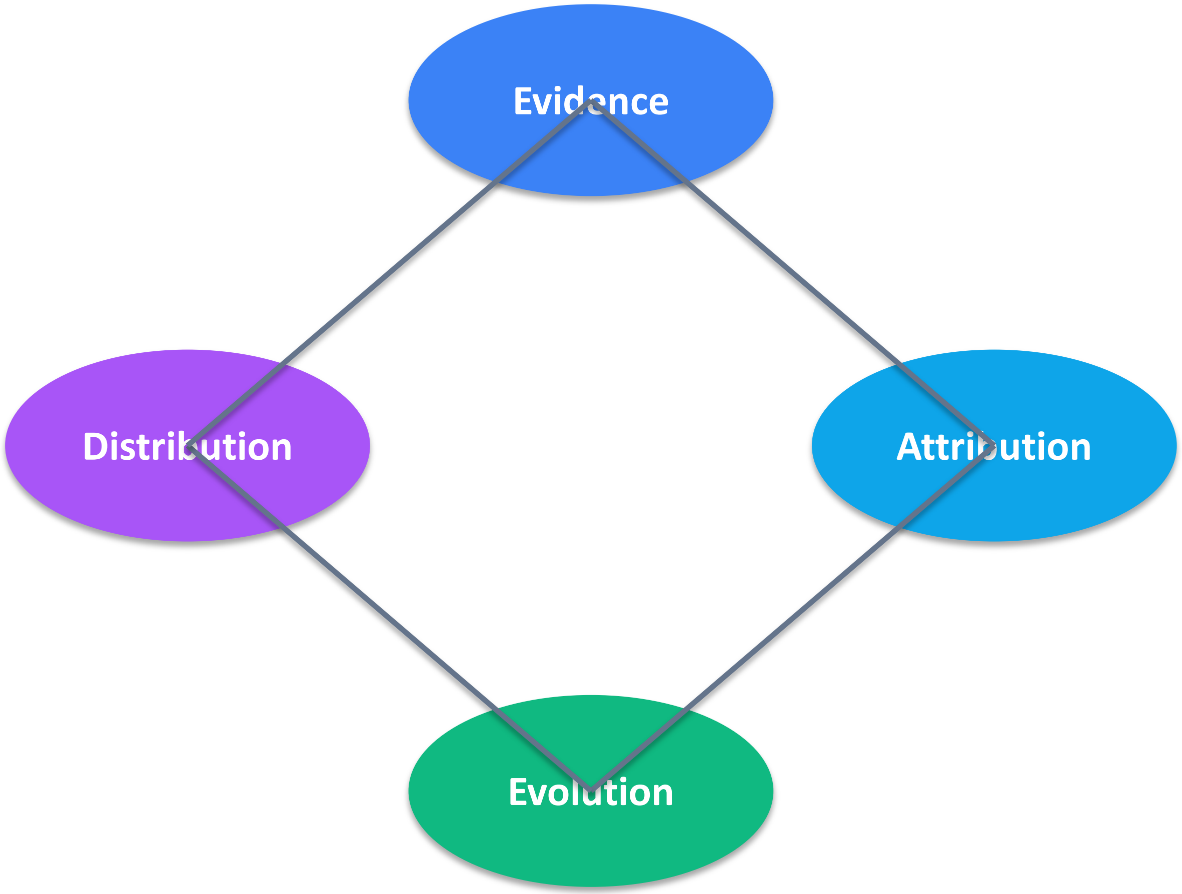
- Skills manually installed
- No runtime learning from trajectories
- Repeated failures across users
- No shared memory of successful fixes

SkillClaw evolving paradigm

- Aggregate cross-user trajectories
- Diagnose root causes with attribution
- Propose and verify skill mutations
- Synchronize accepted updates to all users

Closed-Loop Pipeline: From Multi-User Interaction to Shared Skill Evolutio





Agentic Evolver: Evidence → Attribution → Evolution in Three Stages

1) Evidence analysis

Mine trajectories for success/failure signals and behavior boundaries

2) Attribution

Diagnose root causes: missing instructions, weak constraints, brittle tool-use steps

3) Evolution

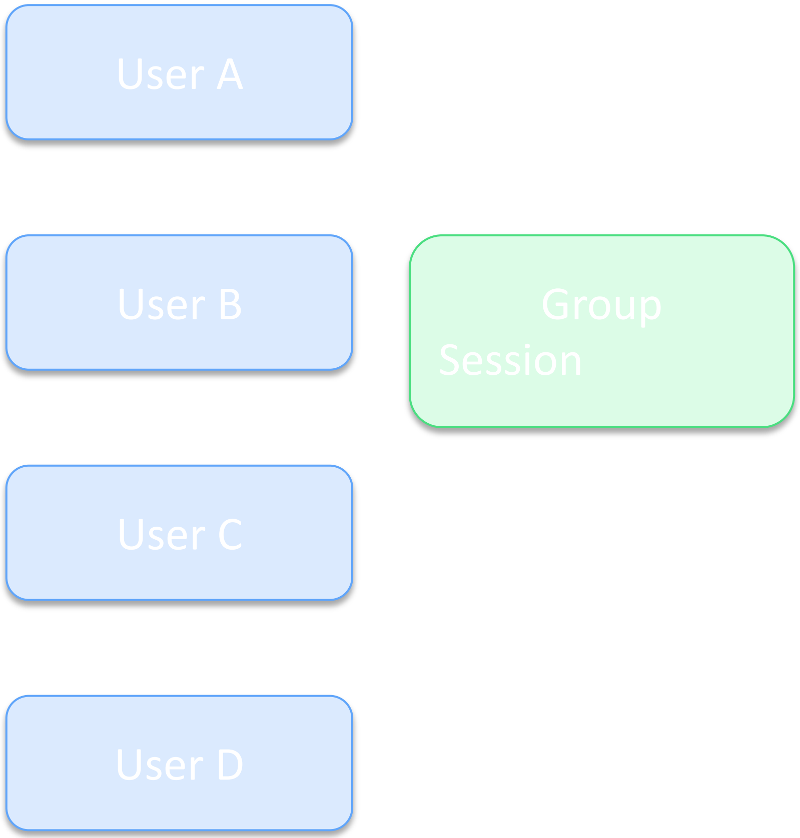
Generate candidate skill mutations through open-ended reasoning and structured edits

Open-ended agent reasoning replaces predefined update rules.

Cross-User Trajectory Aggregation Exposes Skill Behavioral Boundaries

Why multi-user signal matters

- Compatible ecosystems: OpenClaw, CoPaw, IronClaw, PicoClaw, ZeroClaw, NanoClaw, NemoClaw
- Users exercise the same skill under diverse goals, prompts, and environments
- Aggregated trajectories reveal complementary success patterns and failure modes
- Single-user traces are often insufficient to separate generalizable updates from idiosyncratic fixes
- Group session aggregation supplies the evidence base for robust evolution decisions



Case Study: Multimodal Creative Pipeline Skill — 6 Days of Evolution Atte

Date	Attempt	Outcome	Rationale
Day 1	Candidate #1	Rejected	Overfit to narrow prompt style; failed generalization
Day 2	Candidate #2	Rejected	Introduced unnecessary steps; reduced reliability
Day 3	Candidate #3	Accepted	Improved multimodal sequencing without regressions
Day 4	Candidate #4	Rejected	Degraded edge-case handling in tool chaining
Day 5	Candidate #5	Rejected	Validator found inconsistent output formatting
Day 6	Candidate #6	Rejected	No measurable gain over accepted Day 3 version

Result: 1/6 accepted — conservative verification protects skill pool quality.

Case Study: Git Push Auth-Fallback Skill — Iterative Refinement Over 6 Da

Date	Attempt	Outcome	Rationale
Day 1	Candidate #1	Accepted	Added robust auth fallback ordering for common failures
Day 2	Candidate #2	Accepted	Improved credential error diagnosis and retry guidance
Day 3	Candidate #3	Rejected	Extra branches increased complexity without gains
Day 4	Candidate #4	Accepted	Refined non-interactive token path for CI-like contexts
Day 5	Candidate #5	Rejected	Regression in rare SSH-agent configurations
Day 6	Candidate #6	Rejected	Plateau: no improvement beyond accepted revisions

Result: 3/6 accepted — iterative refinement succeeded before plateauing.

Key Takeaways: Three Properties That Distinguish SkillClaw from Existing S

Collective evolution
Individual interactions contribute to a shared, continuously improving skill ecosystem.

Fully automatic
Runtime interaction drives updates without manual curation or explicit user intervention.

Agentic evolution paradigm
Open-ended reasoning over evidence generates context-aware skill mutations.

Implication: build agent ecosystems as continuously learning infrastructures, not static prompt bundles.